

01]:= TableForm[Table[{n, N[vSum[n]], vSum[n]}, {n, 2, 16}]]

01]//TableForm=

2	0.25	$\frac{1}{4}$
3	0.57735	$\frac{1}{\sqrt{3}}$
4	0.957107	$\frac{1}{4} + \frac{1}{\sqrt{2}}$
5	1.37638	$\sqrt{1 + \frac{2}{\sqrt{5}}}$
6	1.82735	$\frac{5}{4} + \frac{1}{\sqrt{3}}$
7	2.30476	$\sqrt[6]{2.30...}$
8	2.80487	$\frac{1}{4} + \frac{1}{\sqrt{2}} + \sqrt{2 + \sqrt{2}}$
9	3.32483	$\sqrt[6]{3.32...}$
10	3.86245	$\frac{1}{4} + \sqrt{5} + \sqrt{1 + \frac{2}{\sqrt{5}}}$
11	4.41597	$\sqrt[10]{4.42...}$
12	4.98395	$\frac{5}{4} + \sqrt{6} + \sqrt{\frac{5}{6} + \sqrt{\frac{2}{3}}}$
13	5.56518	$\frac{1}{2} \operatorname{Csc}\left[\frac{\pi}{13}\right] + \frac{1}{2} \operatorname{Sec}\left[\frac{\pi}{26}\right] + \frac{1}{2} \operatorname{Sec}\left[\frac{3\pi}{26}\right] + \frac{1}{\sqrt{2-2\sin\left[\frac{\pi}{26}\right]}} + \frac{1}{\sqrt{2\left(1+\sin\left[\frac{3\pi}{26}\right]\right)}} + \frac{1}{\sqrt{2-2\sin\left[\frac{5\pi}{26}\right]}}$
14	6.15864	$\sqrt[6]{6.16...}$
15	6.76347	$\sqrt{\frac{46}{3} + \frac{22}{\sqrt{5}} + 8\sqrt{\frac{2}{15}}(25 + 11\sqrt{5})}$
16	7.3789	$\frac{1}{4} + \frac{1}{\sqrt{2}} + \sqrt{1 + \frac{1}{\sqrt{2}}} + \frac{1}{\sqrt{2+\sqrt{2}}} + \frac{1}{\sqrt{2(1+\cos\left[\frac{\pi}{8}\right])}} + \frac{1}{2} \operatorname{Csc}\left[\frac{\pi}{16}\right] + \frac{1}{\sqrt{2-2\sin\left[\frac{\pi}{8}\right]}} + \frac{1}{\sqrt{2(1+\sin\left[\frac{\pi}{8}\right])}}$